



## Problem solving with theoretical probability

- 1 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. Which outcome is most likely to happen? Tick **1** correct answer

|        |   | dice 1 |   |   |    |    |    |
|--------|---|--------|---|---|----|----|----|
|        |   | 1      | 2 | 3 | 4  | 5  | 6  |
| dice 2 | 1 | 2      | 3 | 4 | 5  | 6  | 7  |
|        | 2 | 3      | 4 | 5 | 6  | 7  | 8  |
|        | 3 | 4      | 5 | 6 | 7  | 8  | 9  |
|        | 4 | 5      | 6 | 7 | 8  | 9  | 10 |
|        | 5 | 6      | 7 | 8 | 9  | 10 | 11 |
|        | 6 | 7      | 8 | 9 | 10 | 11 | 12 |

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- 2 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. Which outcomes have a probability of  $\frac{3}{36}$ ? Tick **2** correct answers

|        |   | dice 1 |   |   |    |    |    |
|--------|---|--------|---|---|----|----|----|
|        |   | 1      | 2 | 3 | 4  | 5  | 6  |
| dice 2 | 1 | 2      | 3 | 4 | 5  | 6  | 7  |
|        | 2 | 3      | 4 | 5 | 6  | 7  | 8  |
|        | 3 | 4      | 5 | 6 | 7  | 8  | 9  |
|        | 4 | 5      | 6 | 7 | 8  | 9  | 10 |
|        | 5 | 6      | 7 | 8 | 9  | 10 | 11 |
|        | 6 | 7      | 8 | 9 | 10 | 11 | 12 |

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- 3 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. What is the probability that the dice sum to 5? Tick **1** correct answer

|        |   | dice 1 |   |   |    |    |    |
|--------|---|--------|---|---|----|----|----|
|        |   | 1      | 2 | 3 | 4  | 5  | 6  |
| dice 2 | 1 | 2      | 3 | 4 | 5  | 6  | 7  |
|        | 2 | 3      | 4 | 5 | 6  | 7  | 8  |
|        | 3 | 4      | 5 | 6 | 7  | 8  | 9  |
|        | 4 | 5      | 6 | 7 | 8  | 9  | 10 |
|        | 5 | 6      | 7 | 8 | 9  | 10 | 11 |
|        | 6 | 7      | 8 | 9 | 10 | 11 | 12 |

- ☐  $\frac{1}{6}$
- ☐  $\frac{5}{6}$
- ☐  $\frac{4}{15}$
- ☐  $\frac{4}{36}$
- ☐  $\frac{5}{36}$

- 4 Which calculation would increase 80 by 50%? Tick **1** correct answer

- ☐  $80 \times 0.5$
- ☐  $80 \times 1.5$
- ☐  $80 \times 50$
- ☐  $80 \times 150$

- 5 Increase 20 by 40%. Fill in the blank

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- 6 What is 1.5% of 1000? Fill in the blank

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